INC-4977 — Primary database went read-only for 26 minutes (WAL exhaustion)

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postmortem	sev-1	database	completed

At a glance. On 18 June 2026 the primary Postgres instance for the customer database filled its write-ahead log volume and refused all writes for 26 minutes. The trigger was a replication slot left behind by a decommissioned reporting replica, which held WAL segments that could never be recycled. Reads were unaffected throughout. No data was lost.

Incident record

Jira issue	INC-4977
Related issues	DBA-772, DBA-773, DBA-774, DBA-775, OPS-3290
Severity	SEV-1 (writes unavailable to a tier-1 service)
Environment	Production — pg-cust-prod-01 (primary), pg-cust-prod-02 (standby), PostgreSQL 15.6
Detected at	2026-06-18 21:47 CEST
Mitigated at	2026-06-18 22:13 CEST
Duration	26 minutes
Incident commander	A. Novak
Participants	A. Novak (DBA), D. Fischer (SRE), M. Dubois (on-call), R. Bianchi (duty manager)
Review date	2026-06-24
Status	Closed

Description

At 21:47 CEST on 18 June 2026, every write to the customer database began failing with `PANIC: could not write to file pg_wal/xlogtemp.2891: No space left on device`. PostgreSQL shut down the write path and the instance came back up refusing writes. Registration, profile updates, consent changes and order writes all failed for 26 minutes. Read traffic continued to be served normally from the standby and from the primary itself.

The `pg_wal` volume, provisioned at 200 GiB against a normal steady state of 8–12 GiB, had been growing since 26 May. It crossed 95% at 20:31 and filled at 21:47.

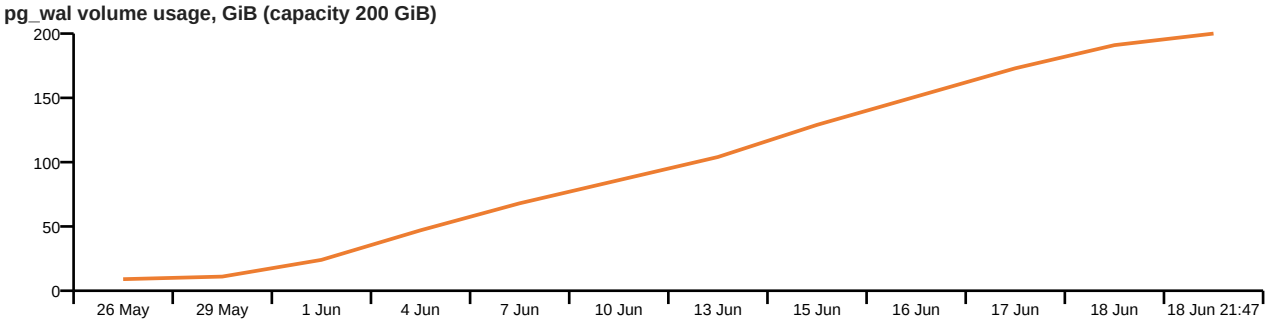

Figure 1 — WAL retained from 26 May, the day the reporting replica was decommissioned. Growth is roughly linear at 8 GiB/day.

Timeline

When (CEST)	What happened
2026-05-26 16:20	Reporting replica pg-cust-rep-03 is decommissioned as part of DBA-701. The VM is deleted. The replication slot `rep_03_slot` is not dropped on the primary.
2026-05-26 onward	The orphaned slot pins WAL. pg_wal begins growing at roughly 8 GiB per day.

When (CEST)	What happened
2026-06-14 03:00	Disk usage crosses 70%. The warning alert fires into the #dba-noise channel, which has notifications muted. Nobody reads it.
2026-06-18 20:31	Usage crosses 95%. The critical alert fires but is configured as a low-urgency notification and does not page.
2026-06-18 21:47	pg_wal reaches 200 GiB. The primary panics and restarts read-only. Application write errors trigger CUST-DB-WRITE-FAIL, which does page.
2026-06-18 21:49	M. Dubois acknowledges and escalates to the DBA on-call.
2026-06-18 21:53	A. Novak joins, declares SEV-1, takes command.
2026-06-18 21:58	`pg_replication_slots` shows rep_03_slot inactive with 191 GiB of retained WAL. Cause identified.
2026-06-18 22:04	Decision recorded on the bridge: drop the slot rather than extend the volume, because the replica no longer exists.
2026-06-18 22:09	`SELECT pg_drop_replication_slot('rep_03_slot');` executed. WAL recycling resumes; usage falls to 12 GiB within 90 seconds.
2026-06-18 22:13	Writes accepted again. Incident mitigated.
2026-06-18 22:45	Queued application retries drained. Incident closed.
2026-06-24 11:00	Postmortem review held with DBA, SRE and the customer platform team.

Root cause

Decommissioning a streaming replica did not include dropping its replication slot on the primary. An inactive slot instructs PostgreSQL to retain every WAL segment the (now non-existent) consumer has not confirmed, indefinitely. WAL therefore accumulated at the rate the database generated it, roughly 8 GiB/day, until the volume was full. The runbook for replica decommissioning, last revised in 2023, listed nine steps and did not mention slots.

Two independent alerts would have caught the growth with days to spare. Both failed for routing reasons rather than for lack of a signal: the 70% warning went to a muted channel, and the 95% critical was classified low-urgency and did not page.

Action plan

Jira	Action	Owner	Due	Status
DBA-772	Add 'drop the replication slot on the primary and verify it is gone' as a mandatory, checked step in the replica decommissioning runbook.	A. Novak	2026-06-26	Done
DBA-773	Alert on any replication slot that is inactive for more than one hour, or that retains more than 20 GiB, paging the DBA on-call.	A. Novak	2026-07-03	Done
DBA-774	Re-classify pg_wal usage above 90% as a paging alert and route it to the DBA PagerDuty service.	D. Fischer	2026-06-30	Done
DBA-775	Audit all production clusters for orphaned slots and remove them. Three further orphans were found on pg-billing-prod-01 and removed.	A. Novak	2026-07-10	Done
OPS-3290	Review every alert routed to #dba-noise; delete or re-route. Muted channels are no longer a permitted alert destination.	D. Fischer	2026-07-17	Done

Follow-up. The three orphaned slots found under DBA-775 had been retaining 4 GiB and 11 GiB respectively; neither cluster was close to exhaustion. All five actions were verified complete at the 2026-07-22 reliability review.